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+ ▾

Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

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0 ▾

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0 forks

0 watching

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3 Branches

0 Tags

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README

A Low-Resource Machine Translation: English To Dholuo

This project builds and evaluates a machine translation pipeline between English and Dholuo, with a focus on adapting the system for medical/telemedicine use.

Problem Statement

Dholuo-speaking patients — particularly rural and elderly patients managing chronic conditions like diabetes or arthritis — often struggle to communicate clearly with doctors on telemedicine platforms. Kenya's doctor-to-population ratio is below WHO guidance, making language-inclusive remote care valuable. No existing Kenyan telemedicine platform (e.g. BYON8) currently supports Dholuo, despite the language having millions of speakers.

Objective:

To build a multilingual English–Dholuo translation model, adapt it toward clinical phrasing, and produce a pipeline that could plug into a telemedicine chat workflow.

Target audience:

Dholuo-speaking patients, Kenyan telemedicine providers, and healthcare workers needing a communication bridge during remote consultations.

Success criteria:

BLEU and chrF score improvement over an untuned pretrained baseline, evaluated on a held-out general-domain test split and a separate medical-phrase test set.

Dataset

Source:

- `1uo_eng.csv` (English–Dholuo parallel corpus, originally from a Kaggle dataset)
- 136,625 sentence pairs, 2 columns (`eng` , `1uo`), no missing values, 11 duplicate rows (dropped)
- English sentences average ~18.3 words (~90 characters); Dholuo sentences average ~19.1 words (~89 characters) — the two languages are closely length-aligned

Project Structure / Workflow

1. **Business Understanding** — problem framing, target audience, success
2. **Data Understanding** — loading and inspecting the parallel corpus
3. **Exploratory Data Analysis**
 - Missing values and duplicates check
 - Sentence-length outlier detection (IQR, boxplots)
 - Class/length balance checks
 - Keyword frequency bar charts and word clouds (English vs. Dholuo)
 - Zipf's law plot (validates the corpus as natural, well-aligned language data)
 - Sentence-length correlation scatter plot (English vs. Dholuo word counts)
 - Vocabulary growth / type-token ratio curve (shows Dholuo's greater morphological richness)
 - Punctuation and special-character frequency comparison (notably the Dholuo apostrophe, which is phonemic rather than auxiliary)

4. Data Preprocessing

- Custom tokenization/cleaning function (markdown/bracket stripping, punctuation-aware)
- Deliberately **no stemming, lemmatization, or stopword removal** — grammatical words (e.g. "not") and verb tense endings are preserved because removing them can invert the meaning of a clinical instruction (e.g. "Do not take this pill" → "Take pill")

5. **Modelling** — train/test split, TF-IDF vectorization, and four progressively more sophisticated translation approaches:

6. **Baseline**: TF-IDF + Nearest Neighbors (retrieval-based)

7. **Seq2Seq (LSTM encoder–decoder)**

8. **Seq2Seq + Attention**

9. **Transformer** (trained from scratch)

10. **NLLB-200 + LoRA** (fine-tuned pretrained multilingual model, hyperparameter-tuned)

11. **Model Evaluation** — BLEU, chrF, validation loss, training time, and parameter count compared across all models

12. Model Explainability

- Cross-attention matrix visualization
- Saliency / gradient-based feature attribution (Inseq)
- Integrated Gradients feature attribution
- Side-by-side comparison scorecard of the explainability methods

8. Final Insights, Recommendations, and Conclusion

Key Results

Model	BLEU	chrF	Val Loss	Training Time
TF-IDF + 1-NN (baseline)	24.50	–	–	–
Seq2Seq (LSTM)	–	–	~5.9–6.0	–
Seq2Seq + Attention	–	33.40	~5.56–5.74	133.5 min
Transformer (from scratch)	9.38	–	3.83	185.0 min
NLLB-200 + LoRA	34.49	50.76	1.17	62.83 min

- **NLLB-200 + LoRA is the clear winner** — highest BLEU/chrF, lowest validation loss, and nearly 3x faster to train than the from-scratch Transformer, demonstrating that transfer learning from a pretrained multilingual model is far more effective than training low-resource NMT architectures from scratch.

Main Takeaways

- Dataset quality (size and diversity), not model architecture, is the primary bottleneck for translation performance.
- The corpus is clean and well-aligned but linguistically low-resource: limited vocabulary diversity and modest lexical growth are visible in the Zipf and type-token analyses.

- Neural approaches substantially outperform the TF-IDF retrieval baseline; attention and self-attention (Transformer) architectures further improve contextual alignment.
- Fine-tuning a pretrained multilingual model (NLLB-200) with LoRA is the most efficient and effective strategy for this low-resource language pair.
- Explainability analyses (cross-attention, saliency, integrated gradients) confirm the fine-tuned model learns genuine linguistic alignments rather than memorizing examples.
- Preprocessing intentionally skips stemming/stopword removal to avoid corrupting clinically important meaning.

Recommendations

1. Expand and diversify the Dholuo–English parallel dataset (books, news, healthcare materials, native speakers).
2. Build a dedicated medical/telemedicine-domain corpus (symptoms, diagnoses, prescriptions, patient-doctor dialogue).
3. Continue leveraging transfer learning from multilingual models (e.g. NLLB-200) rather than training from scratch.
4. Add complementary evaluation metrics (ROUGE, METEOR, BERTScore) and human evaluation.
5. Collect real-world test data from native Dholuo speakers.
6. Extend the pipeline with speech-to-text / text-to-speech for a full voice-based telemedicine workflow.
7. Develop Dholuo-specific tokenizers and normalization rules.
8. Periodically retrain as more data becomes available, and open-source datasets/pipelines to support other low-resource African languages.

Requirements

The notebook was authored to run on **Kaggle with a T4 x2 GPU accelerator** (internet enabled for package installs). Key libraries used:

1. torch
2. transformers
3. peft
4. sentencepiece
5. sacrebleu
6. evaluate
7. inseq
8. pandas
9. numpy
10. scikit-learn
11. nltk
12. matplotlib
13. seaborn
14. wordcloud

How to Run

1. Open the notebook on Kaggle (or another GPU-enabled environment) with the `luo_eng.csv` dataset attached.

2. Ensure the GPU accelerator (T4 x2 or equivalent) is selected and internet access is enabled for `pip install cells`.
3. Run cells sequentially from top to bottom — later sections (Transformer, NLLB-200 + LoRA) depend on artifacts (vocabularies, model checkpoints) produced earlier.
4. Note: one NLLB-200 + LoRA fine-tuning cell is an earlier draft and is intentionally left commented out to avoid duplicate training runs — only the final "Cell 1 – Install Required Packages" NLLB section should be executed.
5. To access translations from deployed model, access via link:
https://huggingface.co/spaces/roy123njuguna/Eng_Luo

Disclaimer

This repository contains a proof of concept pipeline. It is not a validated clinical tool.



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Bennie-Kibet



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Configure

Create and test a Python package on multiple Python versions.



Django

Configure

Build and Test a Django Project



Publish Python Package

Configure

Publish a Python Package to PyPI on release.

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